

Semester: January 2025-April 2025 Maximum Marks: 30 Examination: In-Semester Examination Duration : 1hr. 15 min.		
Programme code: 03 Programme: B.Tech Electronics and Telecommunication Engineering	Class: SY	Semester: IV (SVU 2023)
Institute/School/ Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03C401	Name of the Course: Probability, Statistics and Optimization Techniques	

Question No.		Max. Marks	CO Mapped	BT Level														
Q1a	Find Harmonic mean, mode, median, standard deviation, 1st quartile and 3rd quartile for the following data. Summarize what these measures tell about the data distribution <table><tr><th>Age</th><th>Frequency</th></tr><tr><td>15</td><td>2</td></tr><tr><td>16</td><td>5</td></tr><tr><td>17</td><td>11</td></tr><tr><td>18</td><td>9</td></tr><tr><td>19</td><td>14</td></tr><tr><td>20</td><td>13</td></tr></table> Harmonic Mean (HM): 18.12 Mode: 19 Median: 18.5 standard deviation:1.44 1st quartile and 3rd quartile: 17 and 19 OR Consider two variables: - A=[5, 10, 15, 20, 25] - B=[35, 28, 21, 14,7] Compute the correlation coefficient between A and B. Interpret the result: Does the relationship indicate a strong positive/Negative, weak, or no correlation? The correlation coefficient $r = -1$ indicates a perfect negative correlation. The data shows a strong negative correlation between A and B.	Age	Frequency	15	2	16	5	17	11	18	9	19	14	20	13	05	02	An-Ev
Age	Frequency																	
15	2																	
16	5																	
17	11																	
18	9																	
19	14																	
20	13																	

Q1b	A discrete random variable X has the following probability distribution: <table><tr><td>X</td><td>-3</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>4</td></tr><tr><td>P(X=x)</td><td>0.15</td><td>K</td><td>0.25</td><td>2K</td><td>0.2</td><td>3K</td></tr></table> (a)Find the value of K. (b) Evaluate P(X< 2) and P(−3 < X < 2). (c) Find and plot the cumulative distribution function (CDF) of X. a) 0.15+K+0.25+2K+0.2+3K=1 K=0.4/6= 0.0667 b) P(X<2)=0.15+0.0667+0.25+0.1334 P(X<2)=0.6 P(−3<X<2)=0.0667+0.25+0.1334 P(−3<X<2)=0.45 c) F(x)={0, x<−3 0.15, x =−3 0.2167, x =−1 0.4667, x = 0 0.6, x= 1 0.8, x = 2 1, x≥4	X	-3	-1	0	1	2	4	P(X=x)	0.15	K	0.25	2K	0.2	3K	05	01	An
X	-3	-1	0	1	2	4												
P(X=x)	0.15	K	0.25	2K	0.2	3K												
Q2	A class of 25 students took a physics test, and their scores are as follows: 48, 52, 67, 79, 85, 91, 88, 73, 56, 82, 95, 77, 64, 89, 72, 60, 93, 81, 76, 58, 90, 68, 84, 71, 55 (a) Construct a stem-and-leaf plot for the test scores. (b) Draw a histogram and a pie chart to represent the distribution of scores. Comment on the results (c) Draw Box plot and comment on the results Stem (Tens place) Leaves (Ones place) 2M <table><tr><td>4</td><td>8</td></tr><tr><td>5</td><td>2, 5, 6, 8</td></tr><tr><td>6</td><td>0, 4, 7, 8</td></tr><tr><td>7</td><td>1, 2, 3, 6, 7, 9</td></tr><tr><td>8</td><td>1, 2, 4, 5, 8, 9</td></tr><tr><td>9</td><td>0, 1, 3, 5</td></tr></table> Histogram:2M Pie chart:2M Box Plot: 2M The five-number summary consists of: 1. Minimum (Q0) = 48 (smallest value) 2. First Quartile (Q1) = 67 (25th percentile) 3. Median (Q2) = 76 (50th percentile) 4. Third Quartile (Q3) = 85 (75th percentile) 5. Maximum (Q4) = 95 (largest value)	4	8	5	2, 5, 6, 8	6	0, 4, 7, 8	7	1, 2, 3, 6, 7, 9	8	1, 2, 4, 5, 8, 9	9	0, 1, 3, 5	10	02-03	An-Ev		
4	8																	
5	2, 5, 6, 8																	
6	0, 4, 7, 8																	
7	1, 2, 3, 6, 7, 9																	
8	1, 2, 4, 5, 8, 9																	
9	0, 1, 3, 5																	

	Comment on results: 2M The box plot confirms that the test scores are moderately spread with a slight left skew. Most students performed above average, with a few scoring lower but without significant outliers. This suggests a fairly well-performing class with minimal performance gaps. OR Derive Maximum Likelihood Estimate (MLE) for a Binomial distribution. Calculate the MLE for the mean of a Binomial distribution with observations [0, 1, 1, 0, 1, 1] Derivation: 8M Calculation of MLE for the mean of a Binomial distribution: 2M			
Q3	If V_i, $i = 1, 2, \dots, 20$ are independent noise voltages received in an adder and V is the sum of the voltages received, find the probability that the total incoming voltage exceeds 105 using CLT. Assume that each of the random variable V_i is uniformly distributed over (0,10). uniformly distributed over (0,10). Mean = $\frac{b+a}{2} = \frac{10}{2} = 5$ Var = $\sigma^2 = \frac{(b-a)^2}{12} = 8.33$ $V \sim N(n\mu, \sqrt{n}\sigma)$ $= N(100, 12.9)$ $P(V > 105) = P\left(\frac{V - n\mu}{\sigma\sqrt{n}} > \frac{105 - n\mu}{\sigma\sqrt{n}}\right)$ $= P\left(Z > \frac{105 - 100}{12.9}\right) = P\left(Z > \frac{5}{12.9}\right)$ $= P(Z > 0.387)$ $= 0.5 - 0.1517$ $= 0.3483$	10	03	An